

# FRISBEE

Functional Realtime Intelligence for Spotting Blissful Everyday Actions

Personal Behavioral Alignment Modeling  
Using Logistic Regression on Free-Form Text

Bhavishya · Arya · Hansika · Manogna

Department of Artificial Intelligence & Machine Learning

[github.com/bhavishyasr/frisbee](https://github.com/bhavishyasr/frisbee)

# Abstract

FRISBEE is a personal behavioral modeling system that trains a logistic regression model on an individual's daily free-form journal messages.

The model uses four handcrafted semantic clusters — drift, align, excuse, energy — as TF-IDF weighted features  $x \in \mathbb{R}^4$  to predict *weekly behavioral alignment* (yes-week vs. no-week) as a binary classification task.

## Core Hypothesis

A model trained exclusively on the user's own language outperforms the user's own blinded self-prediction of whether their week aligned with their goals, measured by MAE over a held-out test period.

**Configuration:**  $\lambda = 0.01$  (L2, weights only) ·  $\eta = 0.1$  (learning rate) · Blended IDF (personal + background corpus) · All computation on-device via IndexedDB.

# Introduction: The Identity Gap

There is a persistent, measurable gap between who people *imagine* themselves to be and who actually shows up — day to day.

## Optimism Bias

Morning-you makes promises. Evening-you delivers — or doesn't. The gap is predictable.

## Not a Tracker

Dashboards do not help. Understanding your own language patterns does.

## The ML Question

Can a model learn *your* specific drift patterns better than you predict yourself?

Teenagers experience this most acutely:

*"I said I'd study. I didn't."* *"Why did I do that again?"*

This is not a productivity problem. It is a **self-knowledge problem**. FRISBEE reframes it as an ML problem on *personal* language.

# Objective

1. Build a personal logistic regression model trained *solely* on one individual's daily free-form messages.
2. Predict binary weekly alignment (*yes-week* / *no-week*) from a 4-cluster TF-IDF feature vector in  $\mathbb{R}^4$ .
3. Show that model predictions converge toward ground truth *faster* than the user's own blinded self-predictions, quantified by MAE over a held-out test period.
4. Demonstrate **individual-level** optimism bias modeling — personalized, never population-averaged.

## In Scope

- Single-user, on-device computation (no server, no cloud ML)
- Free-form daily text input (no structured forms)
- Weekly binary outcome label (user-provided ground truth)
- 4 semantic clusters as fixed feature space
- Blended IDF for cold-start handling

## Out of Scope

- Multi-user or cross-user models
- Automated ground-truth labeling
- Real-time or intra-day prediction
- External ML libraries or cloud inference

## Stack (from repository)

TypeScript 96% · React 19 · Vite 7  
Tailwind CSS v4 · IndexedDB (idb)  
Pure-TS ML engine — zero external ML  
deps.

- **Sweller — Cognitive Load Theory**  
Depleted mental capacity causes people to default to low-effort shortcuts, producing poor decision quality.
- **Kahneman (2011) — Thinking Fast & Slow**  
System 1 dominates when System 2 is weakened by fatigue, generating habitual poor choices.
- **Rebetez (2015) — Procrastination & Finance**  
Procrastination routes attention to immediate comfort, systematically delaying corrective action.
- **Chwolka & Rott (2020) — Decision Deferral**  
Choice overload causes students to defer decisions; deferral compounds pressure and degrades quality further.
- **Lee & See (2004) — Trust in Automation**  
Over-trust leads to blind compliance; under-trust causes useful help to be ignored. Both extremes degrade outcomes.
- **Hengstler (2020) — Adaptive Trust Calibration**  
Calibrate trust to match actual system reliability over time — the

# Existing Systems

- **Habit Trackers** (Streaks, Habitica)  
Track completion counts only — no language analysis, no prediction, no personalization of behavioral signal.
- **Journaling Apps** (Day One, Reflectly)  
Store text for recall but run no modeling. Any 'insight' is surface sentiment, not behavioral prediction.
- **Population NLP** (sentiment APIs)  
Models trained on crowd data applied to you. Your patterns are noise, not signal — individual drift is averaged away.
- **Generic Recommenders**  
Suggest behavioral changes with no model of the personal gap between intention and action.

## Common Gap Across All Existing Systems

Built for populations, not persons — and none model the intention-to-action gap from the user's own language.

# Proposed Methodology

1. **Daily Input** User writes a free-form message about how the day went.
2. **Tokenize** Apostrophe-preserving tokenizer (`tokenize.ts`); contractions stay intact.
3. **Feature Extract** 4-cluster TF-IDF scoring (`features.ts`)  $\rightarrow x \in \mathbb{R}^4$  vector.
4. **Predict**  $z = w^\top x + b \rightarrow p = \sigma(z) \rightarrow$  probability of *yes-week*.
5. **Weekly Label** User confirms: *yes-week* or *no-week*?  $\rightarrow y \in \{0, 1\}$
6. **Update**  $w \leftarrow w - \eta[(p - y) \cdot x + \lambda w]; \quad b \leftarrow b - \eta(p - y)$  (L2 on  $w$  only)

*All computation runs on-device in the browser via `engine.ts` + `vault.ts` (IndexedDB). No data leaves the user's machine.*



## Conclusion

### 1. **Personal language is predictive.**

Your daily words encode your behavioral alignment signal:  
drift, alignment, excuses, energy.

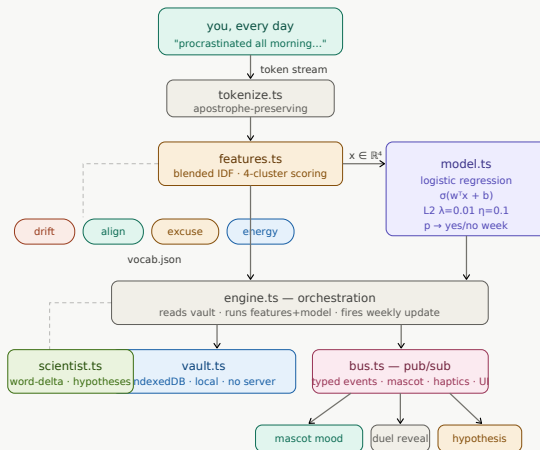
### 2. **Individual $>$ population.**

A model trained only on your data outperforms one  
trained on anyone else's.

### 3. **Optimism bias is modelable.**

The gap between morning-you and evening-you  
is consistent enough to capture in  $\mathbb{R}^4$

# Detailed Architecture



## Step 1 — Feature Engineering

| Cluster | Representative words             |
|---------|----------------------------------|
| drift   | procrastinated, wasted, scrolled |
| align   | focused, proud, accomplished     |
| excuse  | tired, tomorrow, forgot          |
| energy  | motivated, ready, pumped         |

TF-IDF: rare-but-used words carry more signal than frequent ones.

Blended

IDF =  $0.4 \times \text{personal} + 0.6 \times \text{background}$   
(cold-start safe)

## Step 2 — Logistic Regression

**Feature vector:**

$$\mathbf{x} = [x_{\text{drift}}, x_{\text{align}}, x_{\text{excuse}}, x_{\text{energy}}]^T$$

**Linear combination:**  $z = \mathbf{w}^T \mathbf{x} + b$

**Activation:**  $p = \frac{1}{1 + e^{-z}}$

**Loss (binary cross-entropy):**

$$\mathcal{L} = -[y \log p + (1-y) \log(1-p)]$$

**Updates:**

$$\mathbf{w} \leftarrow \mathbf{w} - \eta[(p - y)\mathbf{x} + \lambda \mathbf{w}]$$

$$b \leftarrow b - \eta(p - y)$$

$$\eta = 0.1, \quad \lambda = 0.01, \quad \text{L2 on } \mathbf{w} \text{ only}$$

# Performance Metrics

## MAE — Mean Absolute Error (Primary)

Measures average prediction error vs. ground truth across weeks. Model MAE is compared directly to user self-prediction MAE to validate the core hypothesis.

## Accuracy — Binary Classification

Fraction of weeks where the model's binary decision ( $p > 0.5$ ) matches the user's label  $y \in \{0, 1\}$ .

## Calibration — Confidence vs. Reality

Is  $p = 0.7$  correct  $\approx 70\%$  of the time?  
Evaluated via a reliability diagram over held-out weeks.

## Convergence — Weight Trajectory

How many weeks until weights stabilize?  
Measured by  $\|\Delta w\|$  per update cycle.

*Held-out test period: 4 weeks (4 data points). Results are a proof-of-concept; no statistical significance testing is possible at this scale.*

# Constraints

- **Sample Size**

4-week held-out test = 4 data points. Statistical claims are constrained to proof-of-concept; significance testing is not feasible at this scale.

- **Cold Start**

First 2–3 weeks produce near-random predictions. Blended IDF partially mitigates this but does not eliminate it — the model needs data to be useful.

- **Feature Space**

Fixed 4-cluster vocabulary may miss user-specific language outside the hand-tuned word lists. No

- **Self-Report Bias**

Ground truth  $y$  is user-provided. Recall bias and mood-at-labeling-time introduce noise into the training signal.

- **Single User**

Model is non-transferable. Weights from one person carry no information about another person's drift patterns.

- **Trust Calibration**

Per Lee & See (2004) and Hengstler (2020): early unreliability risks under-trust; overconfident predictions risk over-reliance.

## Future Enhancements

- **Multi-week Test Period**  
Extend held-out evaluation beyond 4 weeks to enable statistically meaningful MAE comparison claims.
- **Dynamic Vocabulary**  
Allow the model to discover user-specific signal words outside the fixed 4 clusters via online feature selection.
- **Temporal Decay**  
Weight recent messages more heavily; behavioral patterns shift over months, so old data should have diminishing influence.
- **Confidence Display**  
Surface the model's uncertainty to the user (per Hengstler 2020) to support adaptive trust calibration rather than blind reliance.
- **Multi-user Study**  
Run FRISBEE across  $N$  participants, aggregate MAE results, and compare optimism bias magnitude between individuals.
- **Cognitive Load Signal**  
Detect Sweller-style cognitive overload from drops in message length and complexity — predict poor decision risk before the week